sys.objects (Transact-SQL)

Applies to: SQL Server · Azure SQL Database · Azure SQL Managed Instance · Azure Synapse Analytics · Analytics Platform System (PDW) · SQL analytics endpoint in Microsoft Fabric · Warehouse in Microsoft Fabric

Contains a row for each user-defined, schema-scoped object that is created within a database, including natively compiled scalar user-defined functions.

For more information, see Scalar User-Defined Functions for In-Memory OLTP.

> **Note**
>
> `sys.objects` doesn't show DDL triggers, because they aren't schema-scoped. All triggers, both DML and DDL, are found in sys.triggers. `sys.triggers` supports a mixture of name-scoping rules for the various kinds of triggers.

Expand table

Column name	Data type	Description
`name`	sysname	Object name.
`object_id`	int	Object identification number. Is unique within a database.
`principal_id`	int	ID of the individual owner, if different from the schema owner. By default, schema-contained objects are owned by the schema owner. However, an alternate owner can be specified by using the `ALTER AUTHORIZATION` statement to change ownership. `NULL` if there's no alternate individual owner. Is `NULL` if the object type is one of the following values: `C` = Check constraint `D` = Default (constraint or stand-alone) `F` = Foreign key constraint `PK` = Primary key constraint `R` = Rule (old-style, stand-alone) `TA` = Assembly (CLR-integration) trigger `TR` = SQL trigger

Column name	Data type	Description
		`UQ` = Unique constraint `EC` = Edge constraint
`schema_id`	**int**	ID of the schema that the object is contained in. Schema-scoped system objects are always contained in the sys or `INFORMATION_SCHEMA` schemas.
`parent_object_id`	**int**	ID of the object to which this object belongs. `0` = Not a child object.
`type`	**char(2)**	Object type: `AF` = Aggregate function (CLR) `C` = Check constraint `D` = Default (constraint or stand-alone) `F` = Foreign key constraint `FN` = SQL scalar function `FS` = Assembly (CLR) scalar-function `FT` = Assembly (CLR) table-valued function `IF` = SQL inline table-valued function (TVF) `IT` = Internal table `P` = SQL stored procedure `PC` = Assembly (CLR) stored-procedure `PG` = Plan guide `PK` = Primary key constraint `R` = Rule (old-style, stand-alone) `RF` = Replication-filter-procedure `S` = System base table `SN` = Synonym `SO` = Sequence object `U` = Table (user-defined) `V` = View **Applies to**: SQL Server 2012 (11.x) and later versions `SQ` = Service queue `TA` = Assembly (CLR) DML trigger `TF` = SQL table-valued-function (TVF) `TR` = SQL DML trigger `TT` = Table type `UQ` = unique constraint `X` = Extended stored procedure **Applies to**: SQL Server 2014 (12.x) and later versions, Azure SQL Database, Azure Synapse Analytics, Analytics Platform System (PDW)

Column name	Data type	Description
		`ST` = Statistics tree **Applies to**: SQL Server 2016 (13.x) and later versions, Azure SQL Database, Azure Synapse Analytics, Analytics Platform System (PDW) `ET` = External table **Applies to**: SQL Server 2017 (14.x) and later versions, Azure SQL Database, Azure Synapse Analytics, Analytics Platform System (PDW) `EC` = Edge constraint

`type_desc`	**nvarchar(60)**	Description of the object type:
		
		`AGGREGATE_FUNCTION`
		`CHECK_CONSTRAINT`
		`CLR_SCALAR_FUNCTION`
		`CLR_STORED_PROCEDURE`
		`CLR_TABLE_VALUED_FUNCTION`
		`CLR_TRIGGER`
		`DEFAULT_CONSTRAINT`
		`EDGE_CONSTRAINT`
		`EXTENDED_STORED_PROCEDURE`
		`FOREIGN_KEY_CONSTRAINT`
		`INTERNAL_TABLE`
		`PLAN_GUIDE`
		`PRIMARY_KEY_CONSTRAINT`
		`REPLICATION_FILTER_PROCEDURE`
		`RULE`
		`SEQUENCE_OBJECT`
		`SERVICE_QUEUE`
		`SQL_INLINE_TABLE_VALUED_FUNCTION`
		`SQL_SCALAR_FUNCTION`
		`SQL_STORED_PROCEDURE`
		`SQL_TABLE_VALUED_FUNCTION`
		`SQL_TRIGGER`
		`SYNONYM`
		`SYSTEM_TABLE`
		`TYPE_TABLE`
		`UNIQUE_CONSTRAINT`
		`USER_TABLE`
		`VIEW`

`create_date`	**datetime**	Date the object was created.

Column name	Data type	Description
`modify_date`	**datetime**	Date the object was last modified by using an `ALTER` statement. If the object is a table or a view, modify_date also changes when an index on the table or view is created or altered.
`is_ms_shipped`	**bit**	Object is created by an internal SQL Server component.
`is_published`	**bit**	Object is published.
`is_schema_published`	**bit**	Only the schema of the object is published.

Remarks

You can apply the OBJECT_ID, OBJECT_NAME, and OBJECTPROPERTY built-in functions to the objects shown in `sys.objects`.

There's a version of this view with the same schema, called sys.system_objects, that shows system objects. Another view, called sys.all_objects, shows both system and user objects. All three catalog views have the same structure.

In this version of SQL Server, an extended index, such as an XML index or spatial index, is considered an internal table in `sys.objects` (`type` is `IT`, and `type_desc` is `INTERNAL_TABLE`). For an extended index:

- `name` is the internal name of the index table
- `parent_object_id` is the `object_id` of the base table
- `is_ms_shipped`, `is_published`, and `is_schema_published` columns are set to `0`

Related system views

Subsets of the objects can be viewed by using system views for a specific type of object, such as:

- sys.tables
- sys.views
- sys.procedures

Permissions

The visibility of the metadata in catalog views is limited to securables that a user either owns, or on which the user was granted some permission. For more information, see Metadata Visibility Configuration.

Examples

A. Return all the objects that were modified in the last N days

Before you run the following query, replace `<database_name>` and `<n_days>` with valid values.

SQL

```
USE <database_name>;
GO

SELECT name AS object_name,
    SCHEMA_NAME(schema_id) AS schema_name,
    type_desc,
    create_date,
    modify_date
FROM sys.objects
WHERE modify_date > GETDATE() - <n_days>
ORDER BY modify_date;
GO
```

B. Return the parameters for a specified stored procedure or function

Before you run the following query, replace `<database_name>` and `<schema_name.object_name>` with valid names.

SQL

```
USE <database_name>;
GO

SELECT SCHEMA_NAME(schema_id) AS schema_name,
    o.name AS object_name,
    o.type_desc,
    p.parameter_id,
    p.name AS parameter_name,
    TYPE_NAME(p.user_type_id) AS parameter_type,
    p.max_length,
    p.precision,
    p.scale,
    p.is_output
FROM sys.objects AS o
INNER JOIN sys.parameters AS p
    ON o.object_id = p.object_id
WHERE o.object_id = OBJECT_ID('<schema_name.object_name>')
ORDER BY schema_name,
    object_name,
```

```
    p.parameter_id;
GO
```

C. Return all the user-defined functions in a database

Before you run the following query, replace `<database_name>` with a valid database name.

SQL

```sql
USE <database_name>;
GO

SELECT name AS function_name,
    SCHEMA_NAME(schema_id) AS schema_name,
    type_desc,
    create_date,
    modify_date
FROM sys.objects
WHERE type_desc LIKE '%FUNCTION%';
GO
```

D. Return the owner of each object in a schema

Before you run the following query, replace all occurrences of `<database_name>` and `<schema_name>` with valid names.

SQL

```sql
USE <database_name>;
GO

SELECT 'OBJECT' AS entity_type,
    USER_NAME(OBJECTPROPERTY(object_id, 'OwnerId')) AS owner_name,
    name
FROM sys.objects
WHERE SCHEMA_NAME(schema_id) = '<schema_name>'

UNION

SELECT 'TYPE' AS entity_type,
    USER_NAME(TYPEPROPERTY(SCHEMA_NAME(schema_id) + '.' + name, 'OwnerId')) AS
owner_name,
    name
FROM sys.types
WHERE SCHEMA_NAME(schema_id) = '<schema_name>'

UNION

SELECT 'XML SCHEMA COLLECTION' AS entity_type,
```

```
    COALESCE(USER_NAME(xsc.principal_id), USER_NAME(s.principal_id)) AS owner_name,
    xsc.name
FROM sys.xml_schema_collections AS xsc
INNER JOIN sys.schemas AS s
    ON s.schema_id = xsc.schema_id
WHERE s.name = '<schema_name>';
GO
```

Related content

- System catalog views (Transact-SQL)
- sys.all_objects (Transact-SQL)
- sys.system_objects (Transact-SQL)
- sys.triggers (Transact-SQL)
- Object Catalog Views (Transact-SQL)
- Querying the SQL Server System Catalog FAQ
- sys.internal_tables (Transact-SQL)

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